Artificial Intelligence-generated Artwork Classification Using Convolutional Neural Networks

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Introduction

People have long been interested in the idea of Artificial Intelligence (AI) being able to generate an image using description, but the realization of such technology has also brought artists to worry over AI Models using artworks from various artists across the internet as part of a large dataset without the artists' permissions which leads to artists having qualms over the users of the AI Model who are being credited for all the work and art it has generated, leaving the artists whose artwork was used to feed the model without proper credit and compensation[1]. With the steady rise of AI technologies, businesses like Midjourney, and DALLE-2 took advantage to advertise their applications with the objective to make art more accessible to those who lack the same technical skills as artists [2].

Such development left many artists with the impression that any advancements in AI-generated artworks can cause a huge impact on the art industry as AI is infamously cheaper than commissioning an artist [3], and with proper prompts, it can even trick people into believing that the AI-generated artwork as a handmade digital artwork and has even won an award in the digital art category.[4]

What truly troubles artists is this cuts out the need to commission artists for their own artworks which affects their own livelihood, and their sources of income as AI art generator usage gains popularity while said generators use other people's artworks to learn how to make their own generated artwork.[5] While even some claim that the AI-generated artwork they posted on various social media platforms is their own creation and has been handmade with their digital tablets and no AI platforms has been utilized.

This paper proposes the classification of AI-generated artworks with the use of Convolutional Neural Networks (CNN) as its main deep-learning architecture through its common features such as colors, the shape of the artwork, and even the consistencies of the overall piece. CNNs can automatically learn features and patterns from images, without requiring manual feature extraction. This is because CNNs are designed to identify spatial patterns in images by applying convolutional filters to the image pixels. These filters detect local features such as edges, textures, and shapes, which are then used to build up higher-level representations of the image.[6]

The use of CNNs for AI artwork classification is still in its early stages, but it has the potential to revolutionize the way that artwork is classified and curated. CNNs can be used to automatically classify artwork, and they can also be used to identify patterns and trends in artwork. The model itself will be utilized to classify if the image being presented is an

AI-generated artwork, it will help both artists and consumers to verify if the artwork is indeed made by human artists or was made with a use of an AI platform. This information can be used to help art historians and curators to better understand and appreciate artwork.

Literature review

Artificial intelligence (AI) has had a significant impact on the art world in recent years. AI-generated artwork, which involves using AI to create images, videos, and other forms of art, has become increasingly popular. [7] The first AI-generated artwork was created in 1973 by Harold Cohen, a British artist. His program AARON could generate drawings based on a set of rules that Cohen programmed into it. In the 1980s, AI-generated artwork began to be used more extensively in computer-aided design (CAD). This allowed for more complex and realistic images to be created. In the 1990s, the entertainment industry began using AI-generated artwork. A famous example would be, the movie "Toy Story" (1995) used AI tools to generate computer-animated characters. This raised awareness of the potential of AI-generated artwork. By the 2010s, AI-generated artwork had entered the art world and was well-received by critics and the public.[4] AI-generated art is still a relatively new field, but it's currently one of the rapidly growing AI trends due to the advances in machine learning and neural networks.

The trend in classifying AI-generated artworks is still new as the technology itself is also still in development and currently improving, and only a few academic researches exist that discuss the method of making a machine learning model that can create a foolproof method in identifying AI-generated artworks.[8]

Currently, there are existing machine learning models that are being made by hobbyists as a proof-of-concept demonstration with the use of Vision Transfomer (ViT) as its main model which is used to predict whether the image being presented is generated by an AI-artwork generator.[9] The website itself is currently deployed in *huggingface.co*, the author stated that the model was trained using a dataset that contains pictures being published on *Reddit*, which resulted in a model that has 80% accuracy, however, the initial results contain several amounts of false positives as most of the images being flagged are actual screenshots and internet memes which as stated not part of the dataset. However, with several adjustments such as the use of AutoTrain, it still has its own limitations with screenshots from the internet, it also has false negatives associated with images generated by Stable Diffusion and DALLE-2. It can also bypass the model by cropping the watermark generated by the AI platform such as DALLE-2, and additional testing revealed that the use of digital image processing filters will trigger false positives. The overall model however is better at differentiating Vector Quantized Generative Adversarial Network (VQGAN) generated images from real paintings.

There is also an AI-generated content detection tool being made by the company Hive Moderation, but it focuses on both textual and image detection, it can provide the percentage if the image is from Stable Diffusion, DALLE-2, and even Midjourney.[10] Hive's API takes an input image and can determine if the image being used is entirely AI-generated or not. The model was trained on a large dataset that contains millions of AI-generated images and

human-made images such as photographs, and digital and traditional artworks. Confidence scores are also provided for each classification of images for efficient interpretation of the results. The Hive's API is generally used to moderate posted artworks and enact bans on AI-generated images.

In the current time, the model itself is still improving, as some images if applied photo filter and image processing glare, the model will declare that the image is AI-generated. Currently, its supported file types are jpeg, webp, and png.

Convolutional neural networks (CNNs) are a type of deep learning algorithm that has been increasingly used for classifying AI-generated artworks. CNNs have several advantages when it comes to classifying AI-generated artworks, including high accuracy where CNNs can achieve high accuracy rates when classifying AI-generated artworks due to their ability to learn hierarchical representations of image features. This is important in the context of AI-generated artworks because they can contain complex and unique visual elements. CNNs are robust to noise and distortions in the input data, which is important in the context of AI-generated artworks since they can have variations in style and structure. It can scale up easily to handle large datasets, which is important when classifying a large number of AI-generated artworks.[11]

However, there are also some limitations to using CNNs in classifying AI-generated artworks, including limited interpretability, making it difficult to understand how the model is making its predictions. CNNs can struggle to generalize to new or unseen AI-generated artworks that differ significantly from the training data. It often requires a large amount of labeled training data to achieve high accuracy rates, which can be a limitation in the context of AI-generated artworks, which may have limited amounts of labeled data available.[12]

Methodology

The best architecture for detecting AI-generated artworks would depend on the specific characteristics of the dataset and the task requirements. The researchers utilized Convolutional Neural Networks (CNNs) for the model's architecture as CNNs are widely used for image analysis tasks and can be effective for detecting AI-generated artworks. They can learn relevant visual features from images and capture patterns that may distinguish AI-generated artworks from traditional human-generated artworks. CNNs can be trained on large image datasets and achieve high accuracy in image classification tasks.

The researchers sampled a dataset that came from Kaggle[13], which contains 3,408 images that are divided into two folders named "FAKE" and "REAL". Each folder contains 1,704 images that resemble post-impressionist artwork, and all of the images have already undergone pre-processing to change to the appropriate size and resolution, making the images uniform and consistent.

The images were divided into parts for training, validation, and testing. For the initial testing, the researchers divided the entirety of the dataset into three equal parts, each folder containing 568 images respectively. The model has shown good performance with the initial experimentation, then the researchers introduced a new partitioning of the dataset. The

researchers decided to feed the majority of the dataset for the training to significantly improve the performance of the model.

To feed the images into the model, the researchers first checked for where the RGB channels are positioned using the c*hannels-first* function. This is important to be able to determine the format of the image. The format that the images fed to the model is important with regard to the model's performance. Next, the researchers then configured the model. The researchers used sequential as the researchers aim to base on a CNN to feed input and expect an output after passing the input through layers.

The model has 4 layers on it. The first layer uses a 2D Convolution Layer (Conv2D) to convolve the image into multiple copies, with regards to the input shape. The next three layers do the same. The use of the ReLU activation is to increase non-linearity in the network. This is because images themselves are highly non-linear; it removes negative values from an activation map by setting them to zero. The researchers then applied the sigmoid activation function to transform the output of the last layer into a probability score between 0 and 1. If the score is near 0, it is most likely fake, and if it nears 1, it is real.

Results

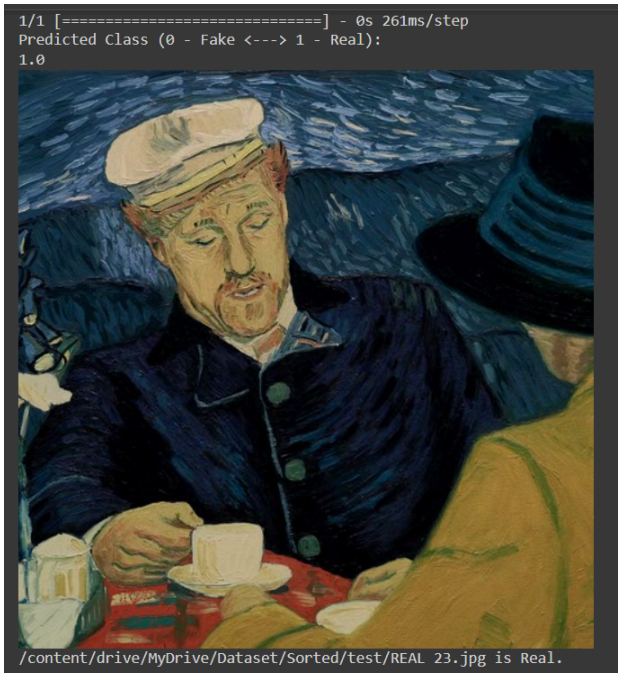


Figure 1.0 *Classification of a Real Image*

The researchers trained the model on a dataset containing 1,704 images of both real and fake, totaling over 3,408 images, combined. The model was trained using 500 samples of training and validation respectively, and over 50 epochs with a batch size of 16. From the first epoch, the model had [loss: 0.4500 - Accuracy: 0.7977 - val_loss: 0.2140 - val_accuracy:

0.9294]. From the first epoch, the model already has a decent accuracy of 79%. The later epochs averaged between 96-98%.

The final accuracy score of the model based on the parameters and the dataset that was fed into it is 98%. After several tests from the test folder in the dataset, the model correctly identified what was fake or real, an example of which is seen in Figure 1.0.

Conclusion

While this model can correctly identify generated and real post-impressionist artwork, it is clear that the model has a long way to go before it can confidently identify AI art that is generated today, as those kinds of art utilize a more advanced AI art generator. Most of the new AI art generators today have been improved to omit the common features that used to be the identifier that art is AI-generated.

This model may be considered mostly as a proof-of-concept as it is tested in a controlled environment, within the bounds of the dataset, and using only one kind of art style, and the research gap on this particular subject is still large. The researchers recommend expanding the types of art, images, and even screenshots to see if it performs well under different circumstances, otherwise, it can try other architectures to see if it can perform better with other types of images and expand the possible applications of the model such as having the model as a general purpose AI-generated image detector.

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